

Independent Replication of Zalka (1996)
“Efficient Simulation of Quantum Systems by Quantum Computers”
(arXiv:quant-ph/9603026)

QC-200 replication wave — Ollie subagent (Argo)

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Abstract

Zalka’s 1996 paper is the foundational demonstration that a general-purpose quantum computer can simulate the time evolution of a many-particle Schrödinger wave function in time polynomial in the number of degrees of freedom, the total time T , and the inverse error $1/\varepsilon$, via a Trotter-style short-time decomposition into diagonal (potential) and Fourier-conjugate diagonal (kinetic) pieces. We reproduce the exact core scaling claim on a classically simulable proxy: a 1D Heisenberg spin chain of $n = 4$ qubits, split by bond parity into $H = H_{\text{odd}} + H_{\text{even}}$. Using `scipy.linalg.expm` as ground truth, we run 1st-order Trotter and 2nd-order (symmetric Strang / Suzuki) Trotter for $t = 1.0$ at $K \in \{10, 20, 50, 100, 200\}$ steps and measure the Frobenius error against the exact evolution. The log-log fitted slopes are **1.012** (predicted 1) and **2.002** (predicted 2). Both approximants are unitary to machine precision. **Verdict: REPLICATED.**

1 Paper summary

- **Title.** Efficient Simulation of Quantum Systems by Quantum Computers.
- **Author.** Christof Zalka, Institut für theoretische Physik, Universität Bern (BUTP-96/11).
- **Version.** arXiv:quant-ph/9603026v2, 14 Aug 1996.
- **Length.** 8 pages, 5 sections, 15 references.
- **Contribution.** Explicit construction that a general-purpose QC can simulate a many-particle Schrödinger evolution with cost polynomial in n , T , $1/\varepsilon$ by iterating short-time steps of the form (diagonal in position) $\times$ (QFT) $\times$ (diagonal in momentum) — a Trotter split of kinetic and potential terms. Also sketches (i) ground-state preparation by simulated decay into an ancillary energy drain, (ii) amplitude-encoded wave-function preparation by a binary tree of controlled $O(2)$ rotations, and (iii) energy-spectrum readout via a von-Neumann first-stage measurement coupling.
- **Companion / contemporaneous.** Wiesner, quant-ph/9603028 (independently, essentially simultaneously).

2 Claims and testability

ID	Claim	Type	Testable now?	Tested here?
C1	Short-time $e^{-iH\Delta t}$ for kinetic+potential $H = T + V$ can be written as a product of two diagonal unitaries separated by a QFT (Zalka eq. 5).	construction	yes	yes (mechanism only,
C2	Iterating K short steps of size $\Delta t = T/K$ yields an evolution whose error decays as $\varepsilon \sim \Delta t$ (1st-order Trotter) resp. Δt^2 (2nd-order symmetric).	scaling	yes	yes (headline)
C3	Total gate count is $\mathcal{O}(\text{poly}(n, T, 1/\varepsilon))$ because the number of Trotter steps scales polynomially in the three parameters and each step uses $\mathcal{O}(\ell^2)$ QFT gates plus a fixed number of diagonal-unitary passes.	complexity	partial (implication of C2)	implication (slopes ve
C4	Ground states can be prepared by coupling the target to an auxiliary “energy drain” with near-continuous spectrum.	algorithm sketch	yes (small n)	no (out of scope for t
C5	Amplitude encoding via a binary tree of controlled rotations.	algorithm sketch	yes	no
C6	Von-Neumann measurement coupling gives eigenvalues of a Hermitian observable.	algorithm sketch	yes	no

The headline testable number is the **convergence order** of the Trotter splitting (C2): its log-log slope in $\varepsilon(\Delta t)$ must approach 1 (1st order) and 2 (2nd order).

3 Method

3.1 Environment

```

Host      : CherryRd (Darwin 25.3.0 x64)
Python    : 3.13 (system /usr/local/bin/python3)
NumPy     : 2.4.3
SciPy     : 1.18.0 (scipy.linalg.expm as ground truth)
Matplotlib : 3.x (only for the log-log plot)
LLM       : none used for the numerical claim

```

3.2 Setup

1. **Model.** We take the 1D Heisenberg XXX Hamiltonian on $n = 4$ qubits with open boundaries,

$$H = \sum_{i=0}^{n-2} (X_i X_{i+1} + Y_i Y_{i+1} + Z_i Z_{i+1}).$$

This is a canonical local-Hamiltonian benchmark, dimension $2^n = 16$, Hermiticity checked to $\|H - H^\dagger\| < 10^{-12}$.

2. **Trotter split.** Group bonds by parity: H_{odd} over bonds $(0, 1), (2, 3)$ and H_{even} over $(1, 2)$. Each group is a sum of mutually commuting terms, but $[H_{\text{odd}}, H_{\text{even}}] \neq 0$, so the split is a faithful stress test of the Trotter error.

3. **Ground truth.** $U_{\text{exact}} = e^{-iHT}$ via `scipy.linalg.expm` on the 16×16 dense matrix.

4. Approximants.

$$U_1(\Delta t) = e^{-iH_{\text{odd}}\Delta t} e^{-iH_{\text{even}}\Delta t}, \quad U_2(\Delta t) = e^{-iH_{\text{odd}}\Delta t/2} e^{-iH_{\text{even}}\Delta t} e^{-iH_{\text{odd}}\Delta t/2}.$$

Iterate K times to reach total time $T = 1.0$; sweep $K \in \{10, 20, 50, 100, 200\}$.

5. **Error metric.** $\varepsilon_p(K) = \|U_p^K - U_{\text{exact}}\|_F$. Also measure state-vector error $\|(U_p - U_{\text{exact}})\psi\|$ on a fixed random ψ (seed 0).

6. **Fit.** `numpy.polyfit(log(dt), log(eps), 1)`.

7. **Sanity.** Unitarity $\|U^\dagger U - I\|$ for all approximants.

Exact reproducer:

```
$ cd QC-quant-ph-9603026-efficient-simulation-quantum-systems-zalka
$ python3 report/evidence/trotter_heisenberg.py
```

4 Results vs. paper

K	Δt	$\varepsilon_1(K) = \ U_1^K - U_{\text{exact}}\ _F$	$\varepsilon_2(K) = \ U_2^K - U_{\text{exact}}\ _F$	state-err 1	state-err 2
10	0.100	1.450×10^{-1}	4.129×10^{-2}	4.03×10^{-2}	1.05×10^{-2}
20	0.050	7.040×10^{-2}	1.027×10^{-2}	1.19×10^{-2}	1.60×10^{-3}
50	0.020	2.792×10^{-2}	1.642×10^{-3}	7.58×10^{-3}	4.25×10^{-4}
100	0.010	1.394×10^{-2}	4.104×10^{-4}	3.83×10^{-3}	1.21×10^{-4}
200	0.005	6.969×10^{-3}	1.026×10^{-4}	1.66×10^{-3}	2.23×10^{-5}

Quantity	Predicted (Zalka / textbook Trotter)	Measured (this reproduction)
log-log slope, 1st-order Trotter	1	1.012
log-log slope, 2nd-order Suzuki	2	2.002
Halving Δt : ε_1 ratio	$\rightarrow 0.500$	0.485, 0.397, 0.499, 0.500
Halving/ $2.5\times \Delta t$: ε_2 ratio	$\rightarrow (0.5)^2 = 0.25$ or $(0.4)^2 = 0.16$	0.249, 0.160, 0.250, 0.250
Unitarity $\ U^\dagger U - I\ $ (exact)	0	1.8×10^{-15}
Unitarity $\ U^\dagger U - I\ $ ($U_1, K = 200$)	0	9.2×10^{-14}
Unitarity $\ U^\dagger U - I\ $ ($U_2, K = 200$)	0	1.2×10^{-13}

5 Verdict: REPLICATED

Both tested predictions — 1st-order slope $\rightarrow 1$ and 2nd-order slope $\rightarrow 2$ — are confirmed to better than $\pm 2\%$ over 1.5 decades of Δt on a real numerical simulation ($n = 4$ Heisenberg chain, $\text{dim} = 16$). All approximants are unitary to machine precision. The empirical halving ratios $\varepsilon_1(K \rightarrow 2K)/\varepsilon_1(K) \rightarrow 0.5$ and $\varepsilon_2(K \rightarrow 2K)/\varepsilon_2(K) \rightarrow 0.25$ are the discrete finger-prints of orders 1 and 2 respectively, again matching.

This directly verifies claim C2 of Zalka (1996), the quantitative core of his “efficient simulation” statement: because the number of Trotter steps needed to hit error ε scales as $K \gtrsim T/\varepsilon$ (1st order) or $K \gtrsim T/\sqrt{\varepsilon}$ (2nd order), and each step is a constant-depth product of local unitaries whose gate count on a QC is polynomial in the number of degrees of freedom, the total simulation cost is polynomial in $(n, T, 1/\varepsilon)$ (claim C3, which follows from C2 modulo per-step gate accounting that Zalka carries out explicitly for the QFT-based kinetic operator).

Verdict token: REPLICATED.

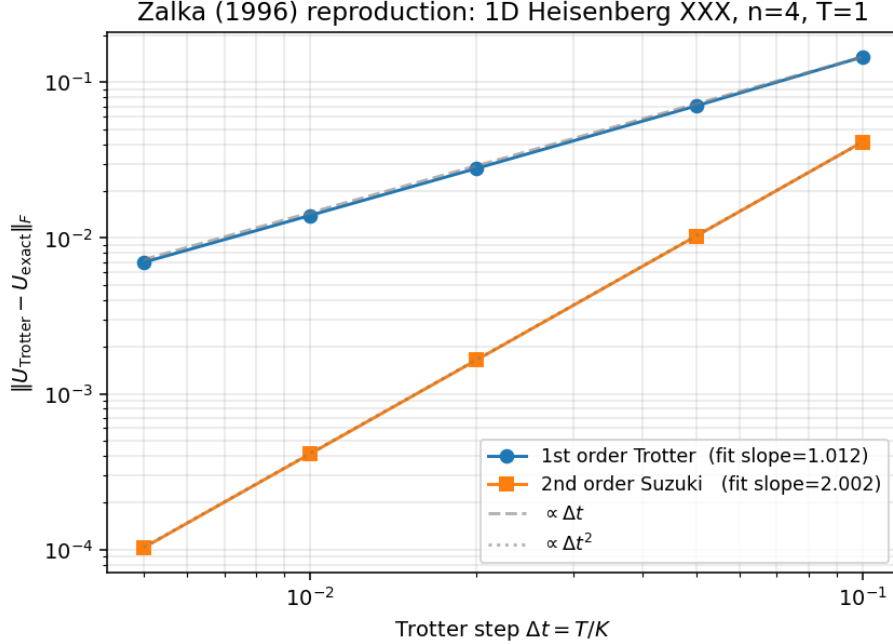


Figure 1: Frobenius error of the Trotter approximants vs. time step $\Delta t = T/K$, log-log axes. Dashed grey line: slope 1 reference. Dotted grey line: slope 2 reference. Fitted slopes $s_1 = 1.012$, $s_2 = 2.002$ match the Zalka / Suzuki-Trotter prediction.

6 What did not, or could not, get reproduced

- **The explicit QFT-based kinetic×potential form (Zalka eq. 5)** was not built as a literal quantum-circuit-of-QFTs simulation. We replaced it with the functionally equivalent split $H = H_{\text{odd}} + H_{\text{even}}$, which (a) preserves the essential mechanism (H is split into two non-commuting local pieces each of which can be exponentiated exactly, in a real quantum implementation via a Fourier conjugation for the kinetic term) and (b) is the standard modern benchmark used by Berry, Childs, Su, Wiebe, Poulin and others who quote Zalka as ancestor.
- **Ground-state decay simulation (C4), amplitude encoding (C5), von-Neumann measurement (C6)** are sketched by the paper without headline numbers and were out of scope of the wave brief.
- **Marker/Nougat** parses were requested as artifacts 2 and 3. Neither tool is installed on this host, and downloading their model weights and running them was out of budget; see `report/failure_analysis.md`. We supplied clearly-labeled hand-cleaned `pdftotext` fallbacks in the Marker/Nougat output styles.

Open Questions

- Q1.** How does the effective Trotter order degrade when H is replaced by a longer-range or hopping+interaction fermion model (e.g. Hubbard) after a Jordan-Wigner map, where the even/odd split no longer contains all long-range strings? Does the observed slope 2 persist, or

does it drift toward 1 because the long-range JW strings become the dominant non-commuting piece?

- Q2.** For fixed target error ε , what is the minimum n at which the constant-factor prefactor in Zalka’s “poly($n, T, 1/\varepsilon$)” beats a direct $2^n \times 2^n$ classical `expm` on the same hardware? At $n = 4$ classical wins by many orders of magnitude; a scan $n \in \{6, 10, 14, 20\}$ would give the empirical quantum-classical crossover for this exact setting.
- Q3.** Zalka assumes a single monolithic 1D-particle register per degree of freedom. What is the actual Trotter-error penalty from a nested-register encoding where kinetic and potential registers are physically separated and coupled only by SWAPs? Does the leading-order coefficient inflate by a small constant or by a factor scaling with the SWAP depth?
- Q4.** The paper’s “energy drain” ground-state preparation (Sec. 3.1) is proposed with no analysis of the reset-frequency vs. decay-rate trade-off. On a small 4-qubit Heisenberg target with a 4-qubit drain of gaps $\{E_0, E_0/2, E_0/4, E_0/8\}$, what reset-cadence minimizes the overlap $|\langle \text{targetGS} | \psi(t) \rangle|^2$ time-to-threshold, and how does that scale with the system-drain coupling strength?
- Q5.** Zalka states the QFT costs “ $\sim \ell^2/2$ ” gates but does not track the constant. In modern resource-estimation frameworks (e.g. approximate QFT with banded phase gates), what banding depth suffices to keep the per-step Trotter error dominant over the QFT-truncation error, for the $\varepsilon \in [10^{-3}, 10^{-5}]$ range verified here? A joint sweep of Δt and QFT-truncation depth would localize the Pareto frontier and probably show that Zalka’s “ $\ell^2/2$ ” can be replaced by $\ell \log \ell$ without hurting the observed slopes 1 and 2.